

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 5

6 Introduction to Computer Architecture (swm11)

Consider the following three correct state machines (**seqA**, **seqB**, **seqC**) written in SystemVerilog.

<pre>module seqA(input clk, input rst, output logic [3:0] a); always_ff @(posedge clk or posedge rst) if(rst) a <= 0; else begin a[0] <= !a[0]; a[1] <= a[0] ^ a[1]; a[2] <= &a[1:0] ^ a[2]; a[3] <= &a[2:0] ^ a[3]; end endmodule</pre>	<pre>module seqB(input clk, input rst, output logic [3:0] b); logic [3:0] n; always_ff @(posedge clk or posedge rst) if(rst) b <= 0; else b <= n; always_comb begin n[0] = !b[0]; n[1] = (b[0] & !n[0]) ^ b[1]; n[2] = (b[1] & !n[1]) ^ b[2]; n[3] = (b[2] & !n[2]) ^ b[3]; end endmodule</pre>
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module seqC(input clk, input rst, output logic [3:0] c);
  logic [15:0] s;
  always_ff @(posedge clk or posedge rst)
    if(rst) s <= 16'd1;
    else    s <= {s[14:0],s[15]};
  always_comb
    begin
      c[0] = s[1] | s[3] | s[5] | s[7] | s[9] | s[11] | s[13] | s[15];
      c[1] = s[2] | s[3] | s[6] | s[7] | s[10] | s[11] | s[14] | s[15];
      c[2] = (|s[7:4]) | (|s[15:12]);
      c[3] = |s[15:8];
    end
endmodule
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Digital system
design

- (a) Why is it important that an implementation of a circuit meets all timing constraints? [2 marks]

Answer: Timing analysis computes the signal paths typically between the output-to-input of DFFs. The clock period must be long enough to allow the longest signal path to propagate its signal to the input of a DFF and guarantee the minimum setup time for that DFF. If the clock period is too short (the clock frequency is too high) then the clocked digital abstraction for the circuit will be broken, i.e. the circuit could fail.

In more detail than is needed: clock jitter and the target maximum circuit temperature are also taken into account.

- (b) What is the complete sequence that each of the three modules (**seqA**, **seqB**, **seqC**) outputs after reset (**rst**) is released? Justify your answer. [6 marks]

Answer: This answer will require some analysis of each state machine to determine its operation. A state transition table might be produced.

Every module produces the following sequence from reset: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 0, 1, ...

- (c) If the three modules were mapped to an FPGA consisting of many 4-input 1-output LUTs (lookup tables), DFFs (D flip-flops) and programmable wiring, what resources would each module require for a minimal implementation? Justify your answer. [6 marks]

Answer:

- **seqA** requires four DFFs to store **a[3:0]** and four LUTs, one for each equation since all equations have four or fewer inputs.
- **seqB** requires four DFFs to store **b[3:0]**. **n[3:0]** is wiring and there needs to be one LUT for each equation since all equations have four or fewer inputs, so four LUTs in total.
- **seqC** requires sixteen DFFs to store **s**. **c[3:0]** is just wiring but each equation requires an eight-input OR gate that requires three LUTs forming a tree of OR gates. So there will be a total of 12 LUTs.

Note for many FPGAs LUTs and DFFs are grouped, so I will accept answers if this is assumed.

- (d) Let us assume that LUTs have an input-to-output delay of $2d$, DFFs have a setup time of d and no other delays, and we ignore wire delays. For each module, what is the minimum clock period in terms of d assuming no clock jitter? Justify your answer. [6 marks]

Answer:

- **seqA** — the input for every DFF comes from just one LUT, so the minimum clock period is $3d$.
 - **seqB** — for a simple implementation preserving the chaining of **n[0]** to **n[1]** to **n[2]** to **n[3]** results in 4 LUTs delays ($8d$) from **b[0]** to **n[3]**, so including the DFF that makes a $9d$ clock period. However, the Boolean optimiser should be able to convert the equation for **n[3]** into the following:
$$\mathbf{n[3]} = (\mathbf{b[2]} \ \& \ !((\mathbf{b[1]} \ \& \ !(\mathbf{b[0]} \wedge \mathbf{b[1]}))) \wedge \mathbf{b[2]}) \wedge \mathbf{b[3]}$$
which can be implemented by one LUT. So after optimisation, the worst case path can be reduced to one LUT and a DFF, i.e. $3d$.
 - **seqC** — the input for every DFF comes directly from the output of another, so in terms of clocking the DFFs, a clock period of d would be sufficient. However, there is logic on the output to complete **c[3:0]** that is two LUTs deep. If we assume that **c[3:0]** will be the input to some other circuit then the minimum delay would be to the input of DFFs, so the longest path would be two LUTs and one DFF, so a $5d$ clock period.
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